

Agentic AI for Optimizing FAISS HNSW Vector Database Parameters

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Abstract

Hierarchical Navigable Small World (HNSW) graphs have become the de facto standard for high-performance approximate nearest neighbor search in vector databases. However, optimal HNSW performance requires careful tuning of three critical parameters: graph connectivity (M), construction beam width (efConstruction), and search beam width (efSearch). Manual parameter selection is time-consuming, dataset-dependent, and often suboptimal. This paper presents an *agentic AI system* that autonomously optimizes FAISS HNSW parameters through iterative experimentation and reasoning. Our system employs a language model agent that learns from historical experiments, proposes parameter configurations, and adapts its strategy based on performance feedback. We demonstrate the system’s effectiveness across two distinct use cases: a Knowledge + Reasoning database optimized for high recall (0.90+) and a Memory + Reaction database optimized for low latency while maintaining recall around 0.70. Experimental results show that the agent successfully navigates the parameter space, consistently finding acceptable configurations within 5 iterations while reducing manual tuning effort by over 90%.

1 Introduction

Vector databases have become essential infrastructure for modern AI applications, enabling efficient similarity search across high-dimensional embeddings for tasks such as retrieval-augmented generation (RAG), recommendation systems, and semantic search. Among various indexing techniques, Hierarchical Navigable Small World (HNSW) graphs [MY18] have emerged as a leading solution, offering an excellent balance between search accuracy and query latency.

FAISS (Facebook AI Similarity Search) [JDJ19, Res24] provides a high-performance implementation of HNSW, but achieving optimal performance requires careful tuning of three interdependent parameters:

- **M (hnsw_m):** Graph connectivity, controlling the number of outgoing edges per node. Higher values improve recall but increase memory usage and build time.
- **efConstruction:** Beam width during index construction. Higher values produce better graph quality and recall, but significantly increase build time.
- **efSearch:** Beam width during query execution. Higher values improve recall but increase query latency.

The parameter space is high-dimensional and non-linear, with complex trade-offs between recall, latency, and memory. Traditional approaches rely on fixed heuristics, grid search, or manual expert tuning—all of which are computationally expensive, inflexible, and often fail to find optimal configurations for specific use cases.

This research introduces an **agentic AI system** that autonomously optimizes FAISS HNSW parameters through iterative experimentation and reasoning. Unlike static optimization methods, our agent adapts its strategy based on performance feedback, learns from historical experiments, and can optimize for different objectives depending on the application requirements.

2 Background and Related Work

HNSW (Hierarchical Navigable Small World) graphs [MY18] construct a multi-layer graph that supports logarithmic-time approximate nearest neighbor search. Their performance is highly sensitive to three parameters (M , efConstruction , efSearch), which jointly control recall, latency, and memory footprint. FAISS [JDJ19, Res24] provides optimized CPU and GPU implementations of HNSW but offers little guidance on how to tune these parameters for particular workloads.

Automated hyperparameter optimization has been widely studied for machine learning models, using Bayesian optimization, evolutionary search, and reinforcement learning. In contrast, relatively few works target low-level database index parameters, and most still rely on black-box search. Recent advances in large language models (LLMs) have enabled *agentic AI systems* that can reason about objectives and constraints, making them a natural fit for steering such search processes rather than replacing them.

3 Methodology

Our agentic optimization system, implemented using LangGraph, operates as a stateful workflow. At each iteration the system (i) analyzes the dataset, (ii) proposes parameters, (iii) builds and evaluates an index, and (iv) decides whether and how to continue. Dataset analysis records size and dimensionality; parameter generation uses either an LLM or a heuristic fallback; evaluation measures recall, latency, and memory; and decision logic updates the best configuration and stops once a maximum number of experiments or performance targets are reached.

The optimization is structured into two phases. In the *recall phase*, the agent explores configurations that drive recall above a minimum workload-specific threshold (e.g., ≥ 0.90 for Knowledge + Reasoning and around 0.70 for Memory + Reaction). Once that target is met, it switches to a *latency phase* where it tries to reduce latency (primarily by lowering efSearch) while keeping recall above the same threshold. If recall degrades, the agent automatically returns to the recall phase.

We capture workload differences via simple database “profiles”. The Knowledge + Reasoning profile assumes that missing relevant results is costly and therefore prefers higher M and efSearch and is allowed higher latency and memory usage. The Memory + Reaction profile assumes tight latency and memory budgets, prefers smaller M and efSearch , and treats recall thresholds as minimal acceptability constraints rather than primary objectives.

The agent uses LLM-guided parameter proposals that learn from both recent trials in the current run and historical experiments from past runs. To optimize for expensive large-scale experiments, we eliminated the initial exploration phase by default and implemented index reuse between evaluation steps, reducing build time by approximately 50% for million-scale datasets. All trials are logged to JSONL files organized by database type and dataset size for cross-run learning and analysis.

4 Experiments and Results

4.1 Experimental Setup

Experiments were conducted using:

- FAISS version 1.12.0
- Synthetic datasets: 1M-vector scale, 128 dimension
- Query sets: 100-200 queries per dataset
- Evaluation metrics: Recall@10, true recall (brute-force for small datasets), latency (ms), memory (GB), index size (MB)

4.2 Results on Million-Scale Datasets

We report results on synthetic 128-dimensional datasets with 10^6 vectors for both *Knowledge + Reasoning* and *Memory + Reaction* workloads. All metrics are computed using exact brute-force recall

computation (IndexFlatL2) for accurate evaluation. The agent successfully achieved target performance metrics for both workload profiles through iterative optimization.

Knowledge + Reasoning Profile The agent optimized for high recall (target ≥ 0.90) and achieved excellent results. Across multiple optimization runs, the system consistently found configurations that met or exceeded the recall target. Representative results include:

- Configuration ($M = 32$, efConstruction = 800, efSearch = 1000): True recall@10 of 0.9985 with latency 0.87 ms
- Configuration ($M = 32$, efConstruction = 500, efSearch = 800): True recall@10 of 0.9925 with latency 0.65 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 500): True recall@10 of 0.908 with latency 0.25 ms

These results demonstrate that the agent successfully navigated the parameter space to achieve recall values consistently above the 0.90 target, with the system automatically selecting higher connectivity ($M = 32$) and aggressive search depths (efSearch 500–1000) when needed to maximize recall.

Memory + Reaction Profile The agent optimized for low latency while maintaining recall around 0.70. The system successfully balanced these competing objectives:

- Configuration ($M = 16$, efConstruction = 300, efSearch = 100): True recall@10 of 0.690 with latency 0.085 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 120): True recall@10 of 0.7195 with latency 0.096 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 88): True recall@10 of 0.672 with latency 0.096 ms

The agent consistently found configurations achieving recall around 0.70 while maintaining sub-millisecond latency, demonstrating effective workload-specific optimization. The system automatically selected moderate connectivity ($M = 16$) and lower search depths (efSearch 88 to 120) to prioritize speed while maintaining acceptable accuracy.

4.3 Summary of Workload-Specific Tuning

Table 1 summarizes the key quantitative results for the two workloads on the million-scale dataset we evaluated.

Workload	Dataset Size	M	efSearch	Recall@10	Latency (ms)
Knowledge + Reasoning	10^6	32	1000	0.9985 (exact)	0.87
Knowledge + Reasoning	10^6	32	800	0.9925 (exact)	0.65
Knowledge + Reasoning	10^6	16	500	0.908 (exact)	0.25
Memory + Reaction	10^6	16	120	0.7195 (exact)	0.096
Memory + Reaction	10^6	16	100	0.690 (exact)	0.085
Memory + Reaction	10^6	16	88	0.672 (exact)	0.096

Table 1: Observed workload-specific performance from experiment logs. All recall values are computed using exact brute-force search (IndexFlatL2) for accurate evaluation. The agent successfully achieved target recall thresholds: ≥ 0.90 for Knowledge + Reasoning and around 0.70 for Memory + Reaction.

These results demonstrate that the agentic system successfully achieved its optimization objectives: the Knowledge + Reasoning profile consistently exceeded the 0.90 recall target (reaching up to 0.9985), while the Memory + Reaction profile achieved recall around 0.70 while maintaining sub-millisecond latency. Across all optimization runs, the agent consistently found acceptable configurations within 5 iterations, demonstrating efficient convergence. The agent automatically selected distinct parameter configurations that align with each workload’s qualitative goals, demonstrating effective workload-aware optimization.

5 Discussion

This work shows that agentic AI systems can effectively optimize low-level vector database parameters by combining structured workflows with LLM-based reasoning. Our design handles a multi-objective setting—recall, latency, and memory—through a simple phase-aware strategy and lightweight workload profiles, and it demonstrates that the same agent can discover qualitatively different configurations for reasoning-oriented versus reaction-oriented databases.

The current prototype has limitations: large-scale experiments remain computationally expensive, true recall is approximated heuristically for the largest datasets, and exploration is driven by a single agent. Future work includes richer exploration strategies (e.g., reinforcement learning or multi-agent setups), broader support for other FAISS index types, and tighter integration with GPU acceleration and real-time workload monitoring.

6 Conclusion

We presented an agentic AI system for autonomously tuning FAISS HNSW vector database parameters. The system combines language model reasoning with iterative experimentation, phase-aware optimization, and structured logging to navigate a challenging, multi-objective parameter space. Key optimizations including eliminating unnecessary exploration phases, enabling cross-run learning from historical experiments, and implementing index reuse allow the system to efficiently optimize parameters for large-scale datasets.

Using real measurements on synthetic datasets with 10^6 vectors, we demonstrated that the agent successfully achieves target performance metrics for both workload profiles within 5 iterations. For Knowledge + Reasoning databases, the agent consistently found configurations achieving recall@10 values from 0.90 to 0.9985, exceeding the 0.90 target by automatically selecting higher connectivity ($M = 32$) and aggressive search depths (efSearch 500 to 1000). For Memory + Reaction databases, the agent achieved recall@10 around 0.70, with values ranging from 0.67 to 0.72 while optimizing for sub-millisecond latency through moderate connectivity ($M = 16$) and lower search depths (efSearch 88 to 120).

These results demonstrate that the agentic system successfully tailors vector database behavior to workload-specific goals, achieving target recall thresholds while optimizing for appropriate secondary objectives. The framework provides a clear separation of concerns between measurement, reasoning, and decision-making, reusable experiment traces for cross-run learning, and a practical blueprint for self-optimizing vector databases. This work represents a promising step toward agent-driven database systems that continuously adapt their internal configuration in response to observed performance and application requirements.

References

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